NICHOLAS G. STILES

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EDUCATION

Massachusetts Institute of Technology, PhD **August 2025-present**

- PhD student in the department of Electrical Engineering and Computer Science
- GPA: 5.0/5.0
- Co-advised by Gregory Wornell and Yury Polyanskiy
- Research interests include information theory, high-dimensional statistical inference, and machine learning
    - Current focus is generalization guarantees for self-supervised representation learning methods

Massachusetts Institute of Technology, BSc **August 2021-May 2025**

- GPA: 4.9/5.0
- Completed Bachelor of Science in Mathematics (Pure) with Minor in Computer Science

Completed Coursework (non-exhaustive)**:**

- Algorithms I & II, Machine Learning, Inference & Information Theory, Algorithms for Inference, Information Theory: Coding and Learning, Fundamentals of Programming
- Random Matrix Theory, Measure-Theoretic Probability, Stochastic Calculus
- Real Analysis, Linear Algebra, Differential Equations, Discrete Mathematics, Abstract Algebra I & II, Functional Analysis, Complex Analysis, Differential Geometry, Topology
- Spring 2026 coursework includes Nonlinear Optimization

PROFESSIONAL EXPERIENCE

NOTE: information is limited due to non-disclosure agreements

Citadel - Quantitative Research Intern, New York, NY **Summer 2024**

- Conducted execution algorithm research within Fixed Income & Macro group
- Applied reinforcement learning methods to improve models of counterparty behavior
- Earned return offer

Massachusetts Institute of Technology – Teaching Assistant, Cambridge, MA **Fall 2023**

- Instructed 25-student class sections twice weekly for advanced undergraduate vector calculus class (18.022)
- Designed practice problems, held office hours, led review sessions, and graded exams

Capital Fund Management - Quantitative Research Intern, New York, NY **Summer 2023**

- Conducted market microstructure research for a systematic hedge fund
- Improved robustness of microstructure models to distribution shift
- Earned return offer

Trexquant Investment - Alpha Research Intern, Stamford, CT **Summer 2022**

- Conducted quantitative research on trading strategies for a systematic hedge fund
- Developed several novel signals which were implemented in the production trading system
- Earned return offer

PROJECTS

Machine Learning for Signal Separation, Deconvolution, and Classification **Fall 2024-present**

- Implemented and compared multiple traditional and modern algorithms for blind adaptive beamforming

- Designed novel transformer model which outperforms current methods while reducing inference time
- Optimized training of large machine learning models using modern GPUs
- Currently working on self-supervised "foundation model" for plug-in application to various tasks

Distribution of Eigenvectors of Random Correlation Matrices **Spring 2024**

- Final project for Random Matrix Theory course taught by Prof. Alan Edelman
- Derived bounds on overlap of empirical and true eigenvectors corresponding to top eigenvalues of multiplicative Wishart perturbation of a fixed correlation matrix
- Used tools from free probability such as the Cauchy transform, S-transform, and resolvent

Top Eigenvalue Process in Dyson's Brownian Motion **Fall 2023**

- Final project for Stochastic Calculus course taught by Prof. Nike Sun
- Showed that the Airy process is the limiting process of the top eigenvalue in Dyson's Brownian motion
- Combined tools from stochastic calculus and random matrix theory, including the Fredholm determinant and the Eynard-Mehta theorem, which characterize the behavior of determinantal point processes

SKILLS, ACTIVIES & INTERESTS

- Mastery of Python and its numerical packages NumPy and Pandas
    - Skilled at building and testing statistical models & hypotheses on real-world datasets
- Extensive experience with PyTorch for training large machine learning models
- Extensive experience with LaTeX typesetting and mathematical writing
- Player for MIT Varsity Lacrosse
    - NEWMAC All-Academic selection from 2023-5, NEWMAC All-Conference selection in 2025
- Interests include theoretical economics and political/moral philosophy, jazz saxophone, weightlifting, skiing